

Exoplanet dynamics

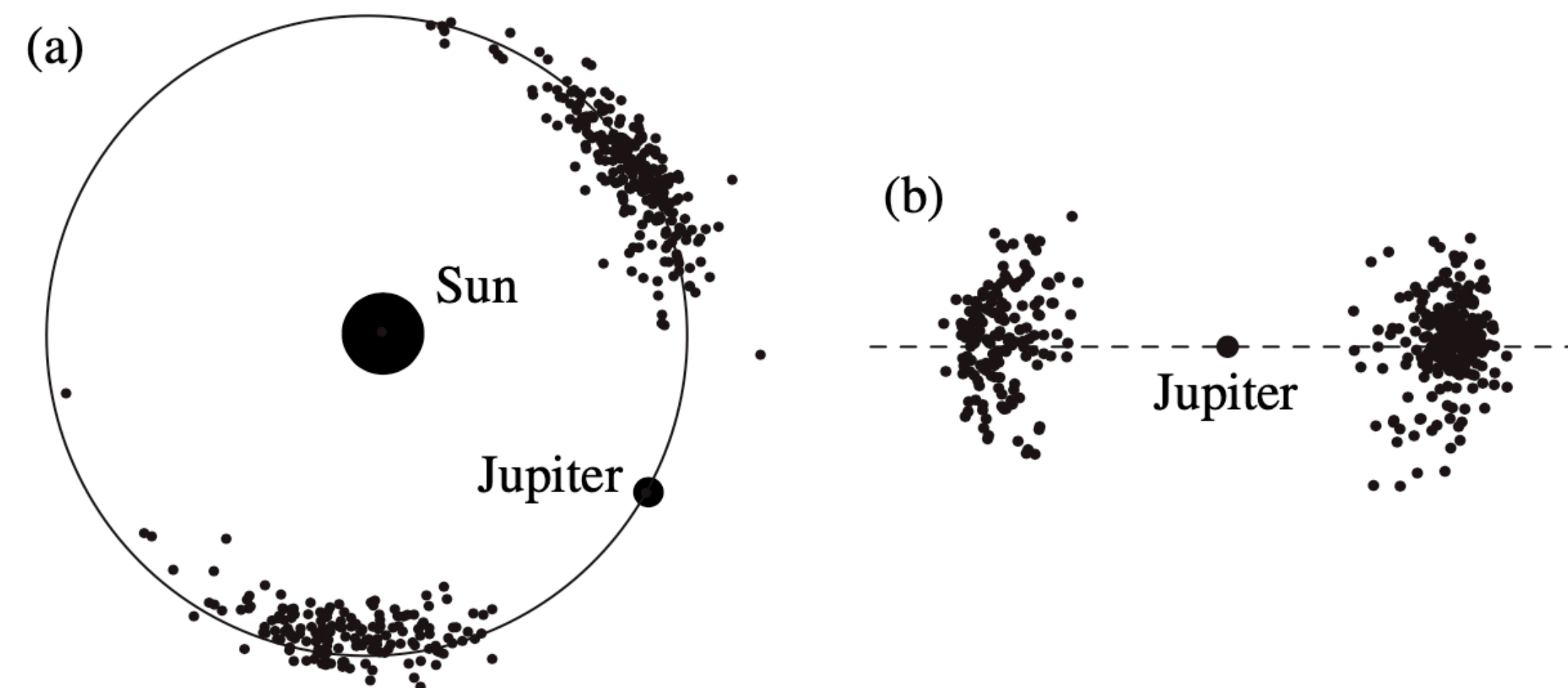
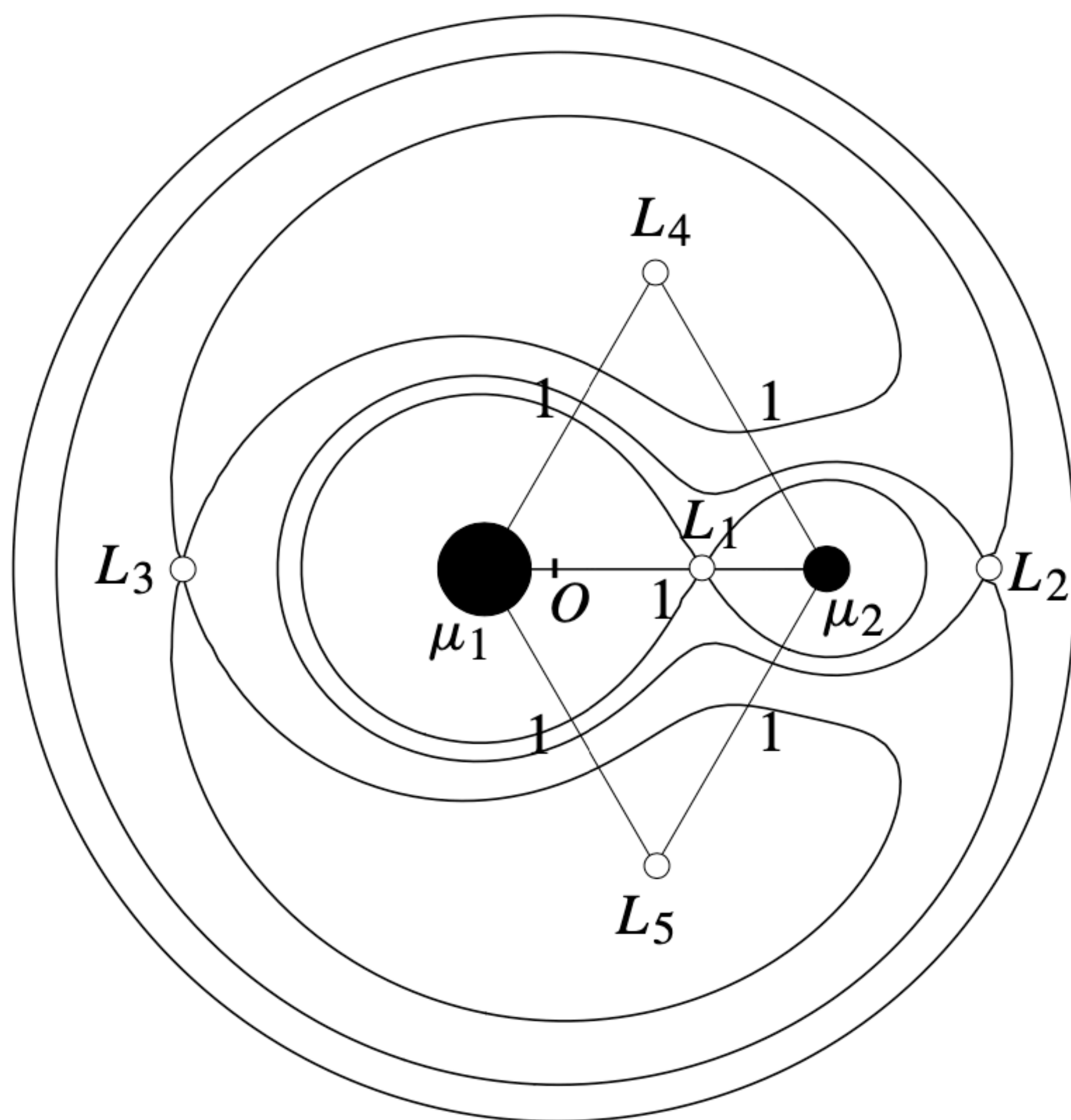
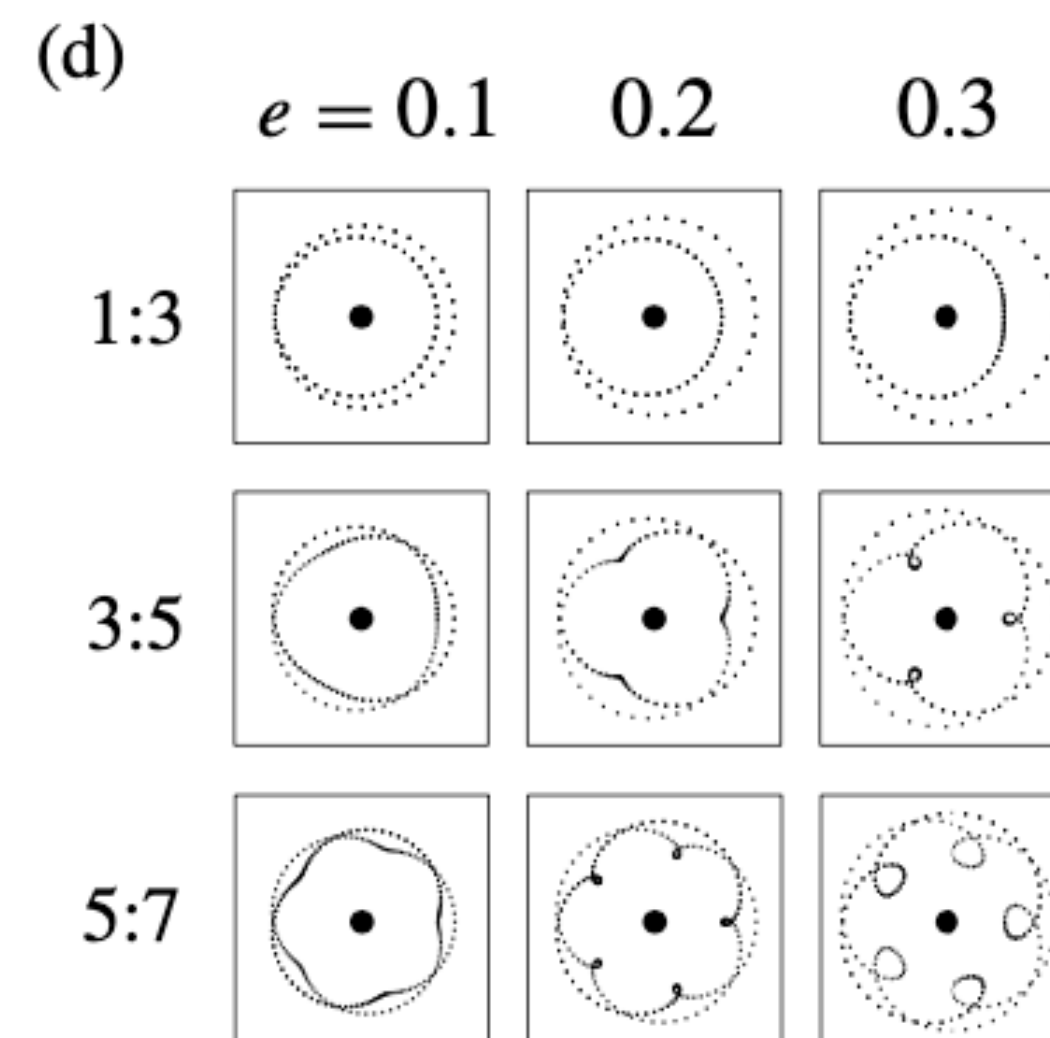
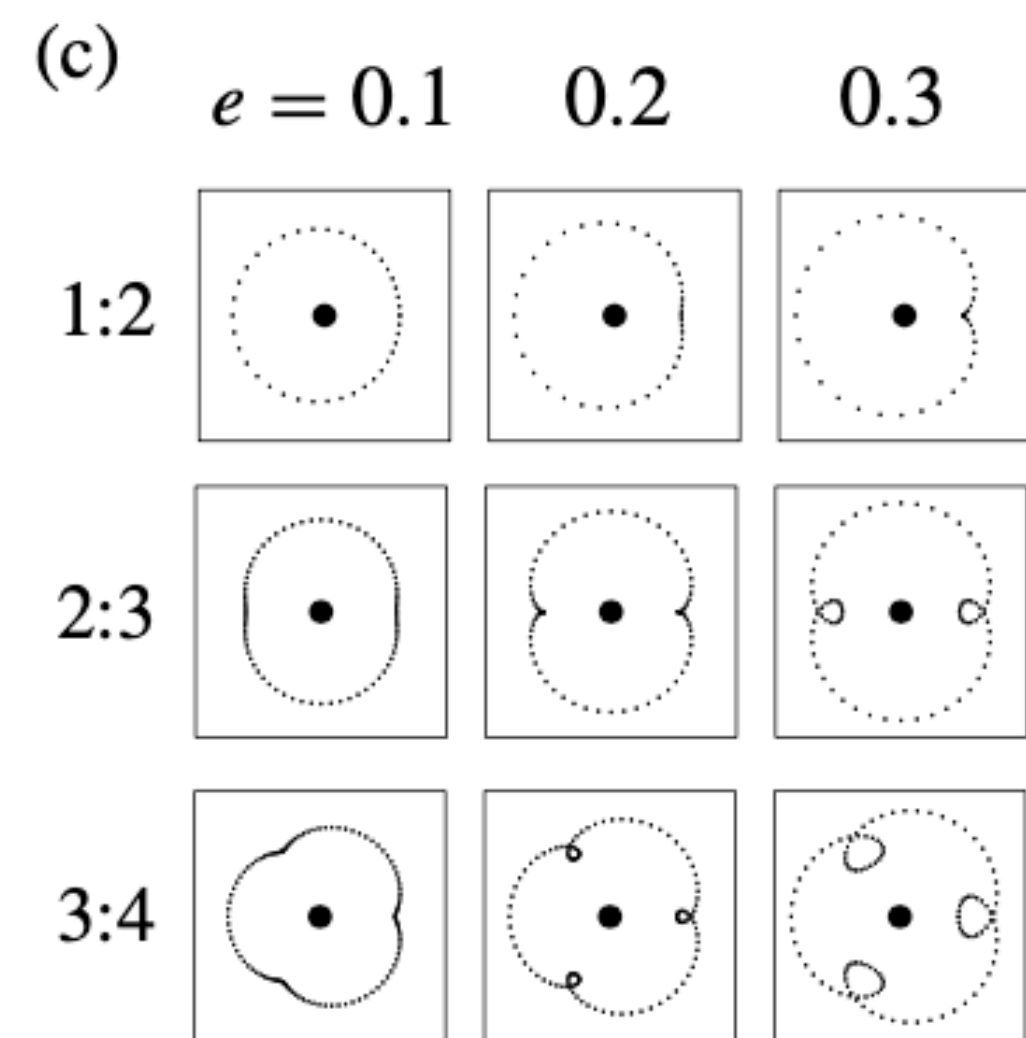
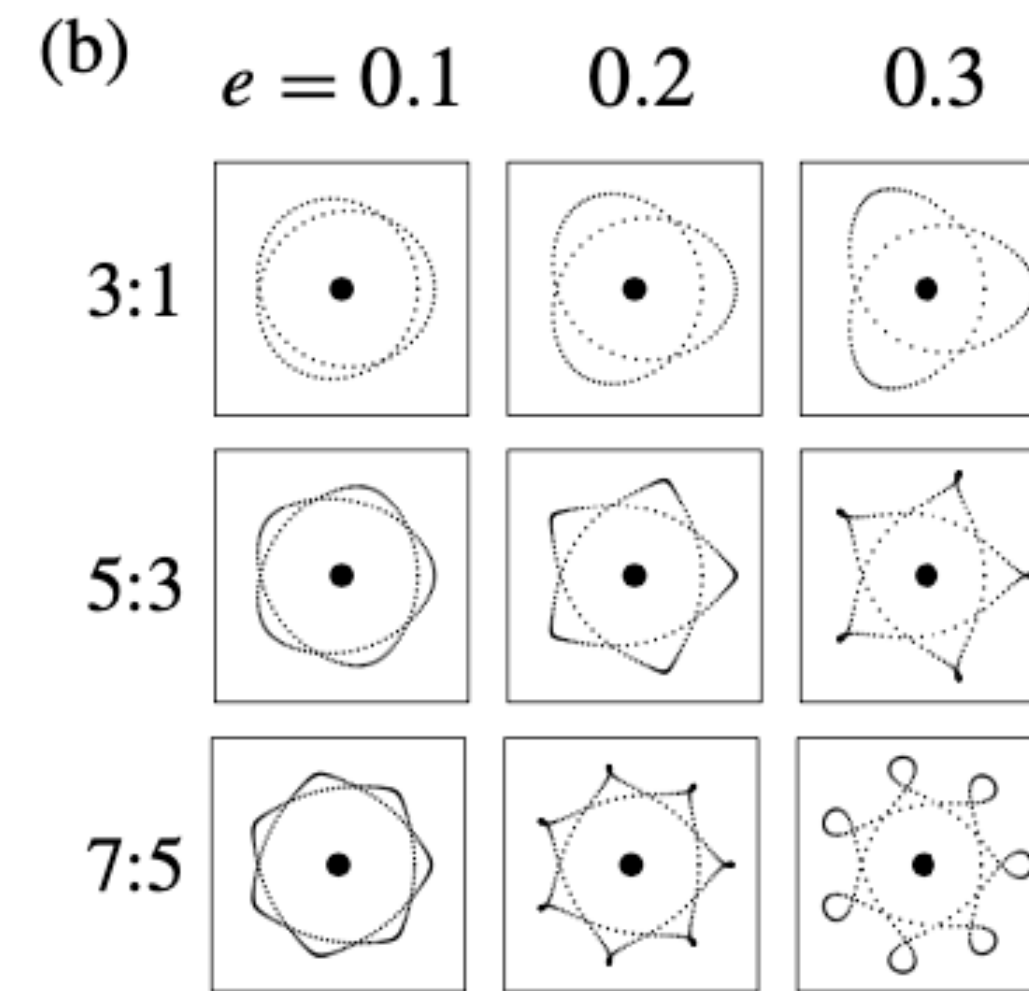
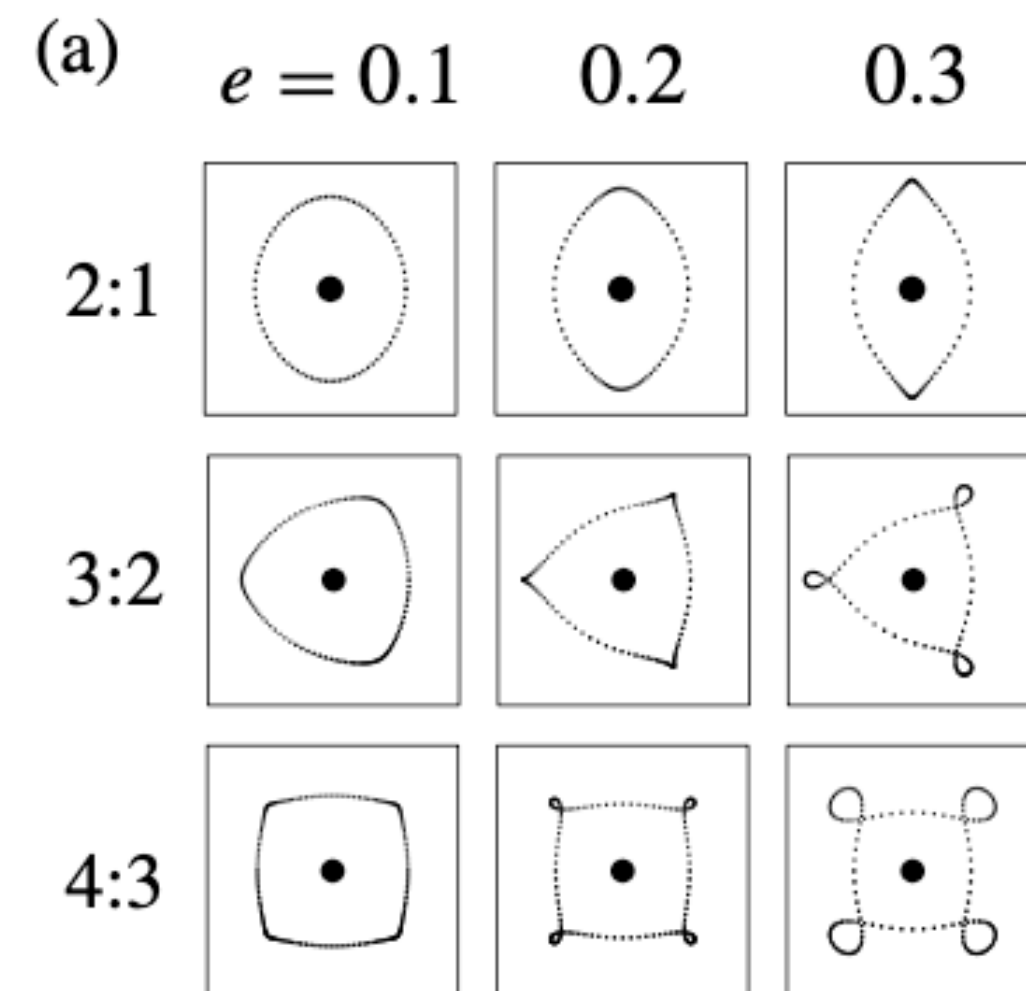


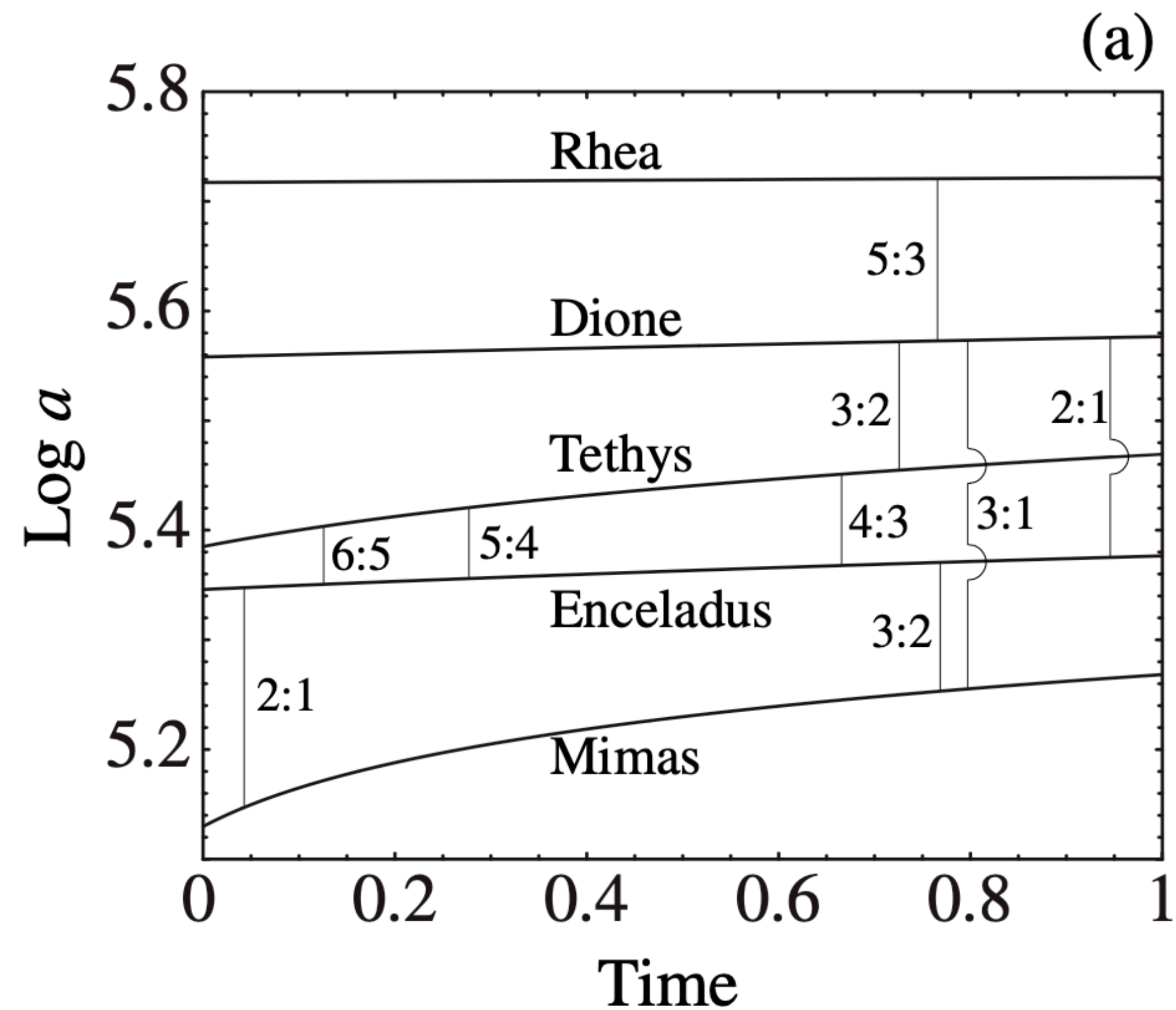
Fig. 3.23. (a) The distribution of asteroids in the vicinity of the orbit of Jupiter on December 18, 1997 at 0^h UT (Julian Date 2450800.5). The plot denotes the positions of the asteroids projected onto the plane of the ecliptic. (b) The vertical distribution of the same asteroids viewed along the Jupiter–Sun line. The dashed line denotes the plane of Jupiter’s orbit.



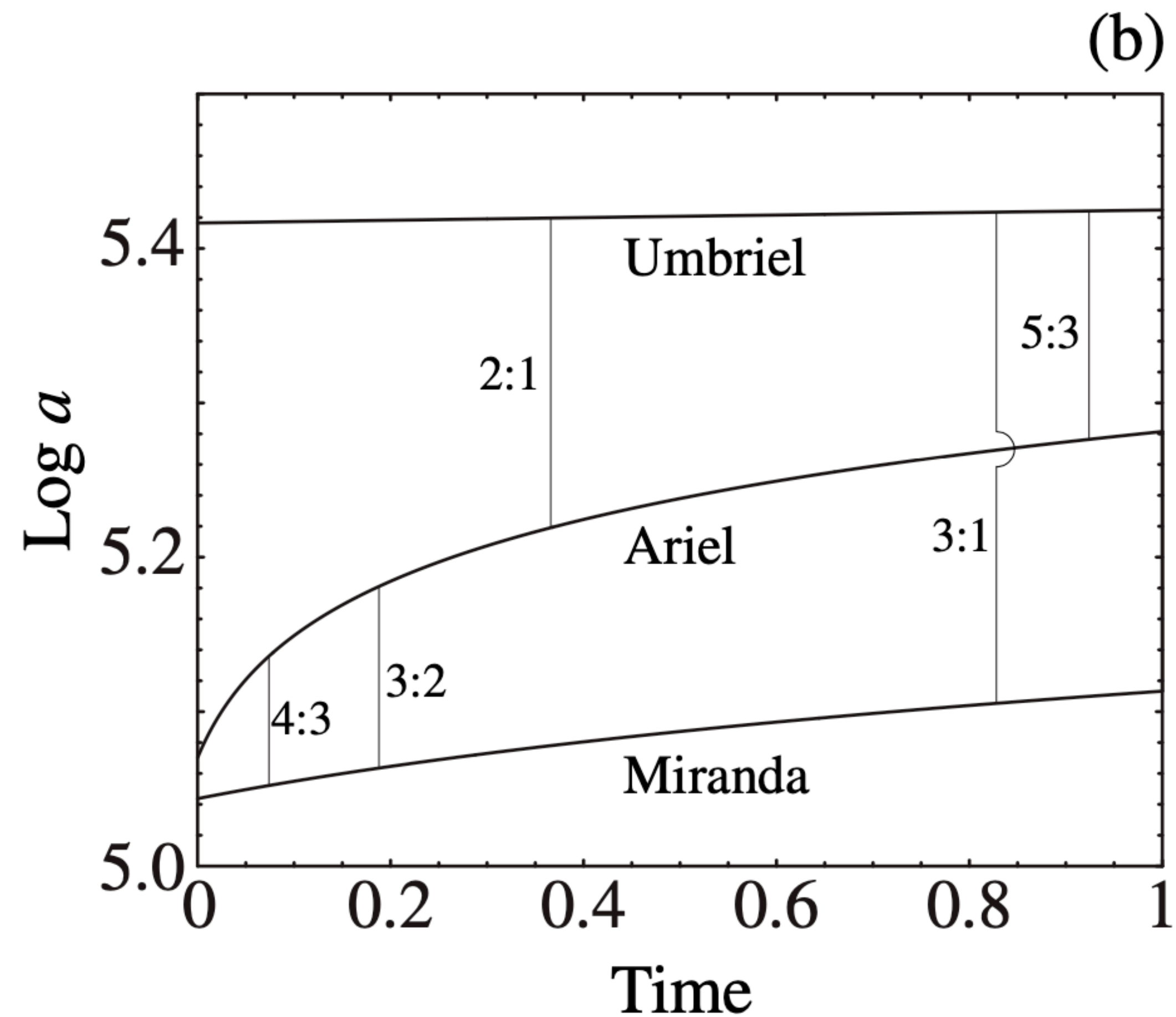
Increased resonance
ratios and eccentricity
cause more
complicated orbits

e is eccentricity of
inner orbit

Saturn's moons



Uranus's moons



Pulsar planets are in a 3:2 resonance

Parameter	Planet 1	Planet 2
m (M_{E})	$3.4 / \sin I$	$2.8 / \sin I$
$a \sin I$ (light ms)	1.31 ± 0.01	1.41 ± 0.01
e	0.022 ± 0.007	0.020 ± 0.006
T (JD)	2448105.3 ± 1.0	2447998.6 ± 1.0
P (s)	5751011.0 ± 800.0	8487388.0 ± 1800.0
ω (deg.)	252 ± 20	107 ± 20